What Impact does Working Out have on Afternoon Energy Levels?

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Experiment Design

Hypothesis: Average energy levels will be higher overall on days where I complete a workout compared to days where no physical activity was completed.

For my experiment I was curious to see if working out at various times was correlated to my energy levels throughout the day and if aerobic (walk/run) or resistance (weightlifting) workouts had more of an impact. I spent two weeks collecting data using a phase based strategy where I spent the first four days doing no physical activity, the next five days I went for a run/walk, and the final five days I did a weightlifting workout at the gym.

Independent Variables:

- **Primary:** Type of Workout (no workout, run/walk, weightlifting)
- Hours of Sleep (prior night's total hours)
- Screen Time (computer and phone)
- Time of Workout
- Length of Workout

Dependent Variable:

- Energy Levels At 10am, 3pm, and 10pm (scale to the right)

Energy Level Scale	
Level	**Description**
1	very low
2	
3	low
4	
5	average, neutral
6	moderate
7	high
8	
9	very high
10	

I was able to follow the experiment fully and complete all the workouts. On some of the off days with no workout I was physically active such as walking to class but no purposeful physical activity was completed. There are a few confounding variables in this experiment that were tracked, such as screen time and hours of sleep, and other variables that were not tracked such as caffeine intake, stress levels and whether or not I took my medication (Adderall). For the untracked variables, I consistently took my meds throughout the whole experiment and only had one cup of coffee each morning, and my stress levels varied throughout the two weeks.

Data Collected

Day	Date	Condition	Hours of Sleep	Screen Time (min)	Workout	Time of Workout	Length of Workout	Energy Level @10am	Energy Level @3pm	Energy Level @10pm
1	10/8	A	6	158	NONE	N/A	N/A	6	4	7
2	10/9	A	7	321	NONE	N/A	N/A	4	5	5
3	10/10	A	7	238	NONE	N/A	N/A	4	3	5
4	10/11	A	10	380	NONE	N/A	N/A	3	5	7
5	10/12	B	6	251	Walk/Run	3:00 PM	54 min	4	8	8
6	10/13	B	10	349	Walk/Run	4:15 PM	40 min	3	6	8
7	10/14	B	8	330	Walk/Run	5:00 PM	47 min	4	4	6
8	10/15	B	8	303	Walk/Run	2:00 PM	45 min	4	6	8
9	10/16	B	8	215	Walk/Run	3:00 PM	40 min	7	8	8
10	10/17	C	10	285	Weightlifting	10:00 AM	55 min	5	4	4
11	10/18	C	8	105	Weightlifting	1:00 PM	60 min	7	8	7
12	10/19	C	8	125	Weightlifting	2:00 PM	50 min	3	6	7
13	10/20	C	8	242	Weightlifting	10:00 AM	65 min	6	4	3
14	10/21	C	8	217	Weightlifting	1:00 PM	48 min	7	5	4

Data Preparation

To prepare the data for analysis I had to do some work on the imported data.

- I removed extra rows from the data set and the energy level scale
- I changed the names of the columns so they were usable (ex. Energy Level @10am, is now energy_level_10am)
- I added a column that turned whether or not I worked out into a binary condition (no workout = 0, any form of workout = 1)
- I added a column that averaged my energy levels for the whole day, taking the average of all three measurements
- I changed screen time and sleep from characters to numeric values

Data Analysis

My process for data analysis involves a series of tests and analysis to examine the hypothesis: does working out affect daily energy levels? The following series of analysis was performed:

1. Summary of Descriptive Statistics
2. Boxplot to visualize differences shown in statistical summary
3. T-test to see if assumptions were statistically significant
4. Correlation to detect and verify confounders
5. Regression to control for confounders
6. Separation of regressions to explore time of day energy levels with box plot

Descriptive statistics:

To get a brief overview of trends in the data and see if my daily energy averages differ between workout and no workout days, I ran a summary looking at mean energy, the standard deviation of energy levels, mean hours of sleep, and mean screen time.

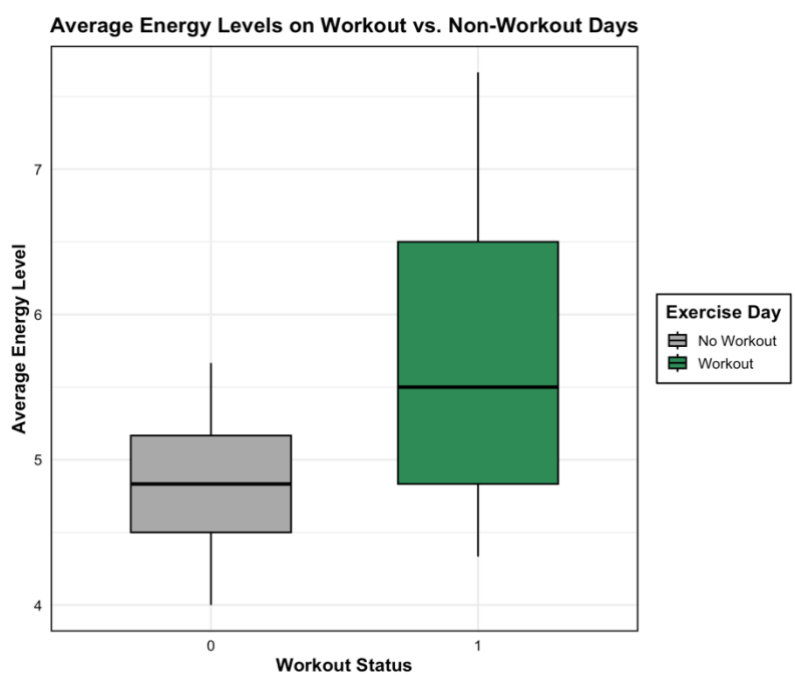
Summary Results:

exercise_day	mean_energy	sd_energy	mean_sleep	mean_screen
0	4.833333	0.6938887	7.5	274.25
1	5.733333	1.1842589	8.2	242.20

Based on the results of these tests it is clear that mean energy is higher on workout days than non-workout days. The standard deviation suggests that my energy levels fluctuate more on days where I workout: this could be contributed to either difference in type of workout or increased energy post workout. Additionally, I got more sleep on days where I was working out and my screen time was less on average. This could have had an impact on my mean energy levels.

Boxplot Visualization:

This visualization of the data highlights that my energy levels are higher on days where I workout with much higher variation. It clearly shows the higher standard deviation on days where I am working out. There are no extreme outliers and the whiskers are longer when I workout indicating higher maximum values. Overall no workout days are more consistently lower energy.



Two Sample t-test:

To confirm that the results indicated in the summary and boxplot are statistically accurate, I ran a two sample t-test.

Results:

- data: avg_energy by exercise_day
- t = -1.4051, df = 12, p-value = 0.1854
- alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
- 95 percent confidence interval:
  - -2.2955934 0.4955934
- sample estimates:
  - mean in group 0 mean in group 1
  - 4.833333 5.733333

Although the summary and visualizations seemed to show a clear increase in energy levels when I worked out, the statistics tell a different story. The mean energy levels are higher on days where I worked out, however the p-value of 0.1854 shows that the difference is not statistically significant.

Correlation to Detect Confounding Variables:

To ensure that my hours of sleep and screen time are not driving the patterns between energy levels and working out I ran a correlation test on both variables.

Correlation Results:

Average Energy and Hours of Sleep

data: df$avg_energy and df$hours_of_sleep

t = -0.61786, df = 12, p-value = 0.5482

alternative hypothesis: true correlation is not equal to 0

95 percent confidence interval:

-0.6459860 0.3914603

sample estimates:

cor: -0.1755892

Average Energy and Screen Time

data: df$avg_energy and df$screen_time_min

t = -1.5315, df = 12, p-value = 0.1516

alternative hypothesis: true correlation is not equal to 0

95 percent confidence interval:

-0.7697805 0.1607084

sample estimates:

cor: -0.4043494

For **hours of sleep**, the correlation coefficient is weak negative. This means there's a slight trend where more sleep might be associated with slightly lower energy but it is so small that it is not meaningful. The p-value confirms this as it is 0.55. The 95% confidence interval includes 0 which further confirms there is no real correlation.

For **screen time**, the correlation coefficient is a moderate negative meaning as screen time increases, energy levels tend to decrease. The p-value is 0.15, closer to the 0.05 cut off but still not statistically significant. The confidence again crosses 0, showing uncertainty about the direction of the effect.

Overall, screen time is more likely to have an effect on energy levels yet it is still not significant. More data is required to fully ensure that this relationship is present.

Regression to Control Possible Confounding Variables:

Although the impact of screen time and hours of sleep are not statistically significant I ran a liner regression accounting for these variables to further clarify results.

Liner Regression Accounting for Screen Time and Sleep Results:

```
Residuals:
    Min      1Q  Median      3Q     Max
-1.42486 -0.83508  0.08401  0.66196  1.80846
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)     6.753747   1.887180   3.579  0.00502 **
exercise_day    0.865700   0.712227   1.215  0.25210
hours_of_sleep  -0.120603   0.284837  -0.423  0.68096
screen_time_min -0.003704   0.004387  -0.844  0.41817
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Residual standard error: 1.093 on 10 degrees of freedom
Multiple R-squared:  0.2713,	Adjusted R-squared:  0.05273
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F-statistic: 1.241 on 3 and 10 DF, p-value: 0.3458

This model used the following equation to calculate these results:

avg_energy = 6.75 + 0.87(exercise_day) − 0.12(hours_of_sleep) − 0.0037(screen_time_min)

This means that on days with exercise, average energy is predicted to be 0.87 points higher than on non-exercise days, while sleep and screen time are constant. This returned a p-value of 0.25 which is not statistically significant, but directionally positive.

Each additional hour of sleep is associated with a 0.12 decrease in average energy leading to a slightly negative relationship. The p-value of 0.68 is not significant and is likely a random relationship

For each additional minute of screen time there is an average energy decrease of 0.0037. Again the p-value of 0.42 shows that this relationship is not statistically significant.

The R^2 value of 0.27 states that the three predictors together explain 27% of the variation in average energy. Overall, none of the predictors definitively explain energy levels although trends suggest exercise may help.

Regressions to Explore Time of Day Energy Levels

Since it is now clear that the potential confounding variables, screen time and sleep are in fact not confounding, I was curious to see the effect on energy levels for each time of day. Again I ran liner regressions on all three time periods, 10am, 3pm, and 10pm in hopes of identifying any smaller impacts of working out that may have gotten lost in the mean values.

Liner Regression Results:

Energy at 10am

Intercept: 4.25

Exercise Day Effect: +0.75

P-Value: 0.43

R-Squared: 0.05

energy_10am = 4.25 + 0.75(exercise_day)

This regression tells us that on non-exercise days my average energy at 10am was 4.25 and on workout days it was 0.75 points higher. The p-value suggests that the relationship is not statistically significant and the r-squared further explains the relationship showing that exercise only explains 5% of variation in my 10am energy levels.

energy_3pm = 4.25 + 1.65(exercise_day)

Energy at 3pm

Intercept: 4.25

Exercise Day Effect: +1.65

P-Value: 0.09

R-Squared: 0.22

This regression shows that again my energy levels at 3pm on days where I don't work out, average to 4.25 and on workout days my energy is on average 1.65 points higher. The p-value is 0.09 showing the highest correlation we have seen so far showing that it is a marginally statistically significant. The r-squared demonstrates that exercising can explain 22% of variation in afternoon energy levels.

energy_10pm = 6.00 + 0.30(exercise_day)

Energy at 10pm

Intercept: 6.00

Exercise Day Effect: +0.30

P-Value: 0.78

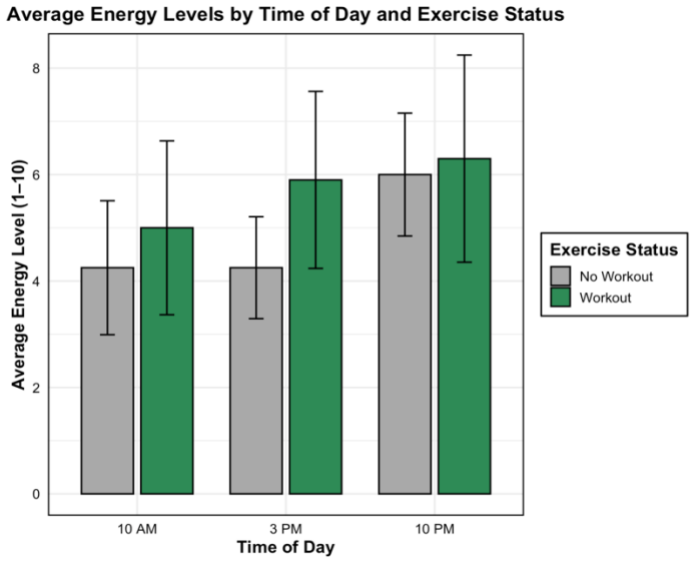
R-Squared: 0.007

Finally, this regression shows that my energy levels at 10pm on non-workout days average to 6.00 and on workout days my energy is about .30 points higher. The p-value of 0.78 suggests that there is no correlation and this is not statistically significant. The r-spared value further highlights this showing that exercise explains less than 1% of variation.

Overall it seems that energy levels are most impacted in the afternoons. This could be explained by post workout endorphins or energy spikes as I tend to workout in the early afternoon. It is understandable that mornings don't see the effects as I did not work out before 10am during this experiment. I believe with more data these correlations would be increased and patterns would develop showing the benefits of working out.

Time of Day Visualization:

This box plot further shows how mid-day energy levels are the most affected by exercising. There are the most high values at 10pm after working out, but that could be actuated to the fact that part of the workout data was collected during the weekend where I was more active at night. Overall working out consistently causes higher energy levels throughout the day.



Conclusion/Reflection

In conclusion, although not statistically significant there is a slight increase in my energy levels when I work out throughout the day with the highest impact in the afternoon. The data, although limited, supported the hypothesis that my average energy levels will be higher overall on days where I complete a workout compared to days where no physical activity was completed.

Limitations:

This experiment had quite a few limitations that caused issues during the data analysis process. First, confounding variables: although analysis concluded that screen time and sleep didn't have an effect on energy levels, there are countless other outside variables that could have had an impact that I could not account for. For example, school work, stress levels, work shifts, overall emotional standing, and menstrual cycle. Secondly, the amount of data: the limited amount of data made it extremely difficult to get a definitive analysis and draw conclusions.

Changes I Would Make:

If I was to test this hypothesis again there are three main things I would change. To start, I would collect data for a much longer period of time. I believe that even four weeks instead of two would allow for a more accurate representation of what my life really looks like and how I feel throughout the day. This would also allow for a more detailed and accurate analysis as it would have more data points to look at. Secondly, I would collect different data. Since it has been made clear that screen time doesn't have too big of an affect I would remove that and focus on things like did I take my medication, caffeine intake, or the food that I'm eating, as these are already scientifically proven to have an effect. Lastly, I would prioritize controls and routine. During the semester my life is all over the place, every day is different, and there are countless outside influences. If I had a set routine or diet that I was following I feel like the data would be more accurate and lead to more definitive conclusions.

What I Learned:

This process taught me a lot about the data collection, analysis, and self-experiments overall. It has lead me to want to do more and follow through with the results of this process. It was relatively easy which I was not expecting and I feel like now that I have gone through the process I will be able to do it again more easily. In the future I would be curious to do more experiments similar to this one now that I have the skills to do so.